Dhanush Reddy Thumma

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.73
Sri Chaitanya Junior Kalasala	Class XII (Telangana State Board of Intermediate Education)	2023	96.60%
SRI CHAITANYA HIGH SCHOOL	Class X (Board of Secondary Education Telangana State, India)	2021	10.00%

WORK EXPERIENCE

Business Analyst Intern, Licious

- **Cannibalization:** Quantified **inward/outward cannibalization** between GUN (mature store) and the nearby new BLG using first-ever-offline-store attribution; **45% of BLG's new customers had already shopped at GUN**, only **3%** of BLG-acquired customers later shopped at GUN, informing GUN win-back action.
- **Catchment revenue model:** Built a **LightGBM model** to predict revenue for catchments with no sales history (Blinkit/Instamart/Zepto dark-store density, demographics, restaurants, real estate); Improved **R squared from 0.245** to **0.401, WAPE 30.4%**, with city-wise reliability shared (Chennai 11% to Kolkata 44%).
- **Pricing experiment (Borewell):** Ran a **local-price-matching test** at the new Borewell store (vs premium pricing) using **pre/post and test-vs-control**; found new-store ramp-up confounded the price effect and **recommended re-running on an established store**
- **CTO incrementality:** Built a **7-bucket customer segmentation** on channel history to isolate true incremental revenue of Call to Order (CTO) from walk-in shifting; ran monthly **MoM and pre/post analysis** on transactors, AOV, ARPU and 30/90-day lapse per bucket.
- **Retention funnels:** Built the **MCM cohort retention funnel** (% retained by acquisition month) and an **O1-O5 order-progression funnel** (time and drop-off from 1st to 5th order across 7/14/28/56/91-day windows) for CRM targeting.

PROJECTS

A/B Test: Level-Gate Placement in a Mobile Game (Cookie Cats)

- Analysed a randomized experiment on **90K players** (gate at level 30 vs 40) using **D1/D7 retention** as primary metrics and **rounds played** as a guardrail; ran a **sample-ratio check** (minor imbalance flagged) and applied **Holm correction** across metrics.
- Found the later gate **cut D7 retention from 19.0% to 18.2%** (**-0.8pp, p = 0.003, 95% bootstrap CI [-1.3, -0.3]**), with **no significant D1 or rounds-played change**; recommended **keeping the gate at level 30**.

Coal Demand Forecasting for Indian Thermal Power Plants | Research Project

- Scraped and parsed 215K+ plant-day records from CEA's Daily Coal Report; ran demographic analysis across 190 thermal plants finding stockout risk scales monotonically with supply chain length (pithead 4.9% critical vs sea-supplied 38.5%)
- Applied k-Shape and DTW time series clustering; formally established via Hopkins statistic that the fleet is a behavioural continuum, then built a global LSTM with learned per-plant embeddings as the forecasting architecture
- Targeting journal publication; continuum finding and embedding-based forecasting constitute the paper's central contribution

Generative Adversarial Network(GAN)-based EMNIST Letters Generator

- Developed a **Generative Adversarial Network(GAN)** that generates realistic handwritten **English letters (A-Z)** using the **EMNIST Letters** dataset.
- Implemented a **Deep Convolutional GAN (DCGAN)** architecture in **TensorFlow / keras,** where the **Generator** synthesizes letter images and the **Discriminator** distinguishes them from the real samples.
- Trained the model over multiple **epochs** with adversarial optimization using **Adam optimizer** and **binary cross-entropy loss,** achieving visually convincing letter outputs.
- Integrated efficient data handling through **TensorFlow Datasets (TFDS)** with normalization and batching.
- Achieved **~90% discriminator accuracy** using a custom **CNN model**, in differentiating real vs generated samples, demonstrating effective adversarial learning.

Variational Autoencoder (VAE) Implementation for Generative Modeling

- Implemented a **Variational Autoencoder (VAE)** to learn latent representations and generate realistic synthetic data.
- Built an **encoder-decoder architecture** leveraging **probabilistic latent variables** for structured and continuous latent space learning.
- Optimized training using **Evidence Lower Bound (ELBO)** with **KL Divergence regularization** to balance reconstruction and distribution learning.
- Evaluated model efficiency through **latent space visualization**, **reconstruction loss**, and qualitative comparison of generated outputs.
- Utilized **TensorFlow/Keras** for model design, training, and experimentation; gained hands-on experience with **generative AI** and **deep probabilistic models**.

RELEVANT COURSES

Online Courses

Machine Learning by Andrew NG - Coursera : Supervised and Unsupervised Learning, Advanced Learning Algorithms, Recommendation system and Reinforcement Learning

Courses on campus

- **Computer Science** : Introduction to Programming
- M**athematics** : Applied Mathematics in Chemical Engineering, Numerical Methods, Elementary Linear Algebra, Calculus

SKILLS

- **Languages and Querying**: SQL(Trino/Presto), Python, C++
- **Python & ML Libraries**: Pandas, NumPy, scikit-learn, Matplotlib, TensorFlow

POSITIONS OF RESPONSIBILITY

E-Cell IIT Hyderabad

Marketing Manager *(May 2024 - May 2025)*

- Led the Cafe Catchup Sessions -- founder-interaction events pairing invited startup founders with ~15 students per session across Hyderabad colleges; personally ran Hyderabad sessions and scaled the format to other cities (e.g., Bangalore) by recruiting and managing city-level POCs under the E-Cell brand
- Conducted E-Summit'25 in IITH, drawing participation from 50+ startups

Media & Publicity Assistant Manager *(Sep 2023 - May 2024)*

- Edited videos and designed mail templates for E-Cell's social media and outreach, supporting event promotion and audience engagement

EXTRACURRICULAR

- **Video Editing** : Freelanced as a video editor helping clients gain thousands of views on various social media
- **Athletics** : Member of NSO; represented my department at **Diesta**, IITH's inter-departmental fest.
- **Sports Enthusiast** : Active follower of **football** and **Formula 1**; regularly participate in campus football matches